



DATABASE NORMALIZATION



OVERVIEW

- What is Normalization?
- 1st Normal Form
- 2nd Normal Form
- 3rd Normal Form

WHAT IS NORMALIZATION?

- **Normalization** is the process of minimizing redundancy from a relation or set of relations.
- Unnecessary redundancy increases the size of databases and can cause problems if there are inconsistencies
- Normalization helps keep our databases from having redundant data and from getting too large
- **Normal forms** are used to eliminate or reduce redundancy in our database tables

1ST NORMAL FORM

- A table is not considered to be in First Normal Form (1NF) if a single cell contains composite or multi-valued attributes
- Meaning if you have a phone number cell with two phone numbers in it, the table violates 1NF and is NOT considered normalized
- A table is considered to be in 1NF if every attribute is a single valued attribute

1ST NORMAL FORM EXAMPLE

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY
1	RAM	9716271721, 9871717178	HARYANA	INDIA
2	RAM	9898297281	PUNJAB	INDIA
3	SURESH		PUNJAB	INDIA

Table 1

Conversion to first normal form

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY
1	RAM	9716271721	HARYANA	INDIA
1	RAM	9871717178	HARYANA	INDIA
2	RAM	9898297281	PUNJAB	INDIA
3	SURESH		PUNJAB	INDIA

Table 2

2ND NORMAL FORM

- For a table to be considered 2nd Normal Form (2NF):
 - The table must be in 1NF
 - Relation does not contain any partial dependency
- A partial dependency occurs when a non-prime attribute is functionally dependent on part of a candidate key
- In other words, every non-key attribute is fully dependent on the primary key

2ND NORMAL FORM EXAMPLE

Unnormalized

<StudentProject>

StudentID	ProjectID	StudentName	ProjectName
S89	P09	Olivia	Geo Location
S76	P07	Jacob	Cluster Exploration
S56	P03	Ava	IoT Devices
S92	P05	Alexandra	Cloud Deployment



Normalization (Second Normal Form)

<StudentInfo>

StudentID	ProjectID	StudentName
S89	P09	Olivia
S76	P07	Jacob
S56	P03	Ava
S92	P05	Alexandra

<ProjectInfo>

ProjectID	ProjectName
P09	Geo Location
P07	Cluster Exploration
P03	IoT Devices
P05	Cloud Deployment

3RD NORMAL FORM

- A table is considered to be in third normal form (3NF) if:
 - there is no transitive dependency for non-prime attributes
 - the table is already in 2NF
- A transitive dependency in a database is an indirect relationship between values in the same table that causes a functional dependency
 - If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$ is a transitive dependency.
- Transitive dependencies can create delete, update, and insert anomalies in the database

3RD NORMAL FORM EXAMPLE

Author_ID	Author	Author_Nationality
A1	Andrew	USA
A2	Andres	USA
A3	Rodney	Canada

- The table above is not in 3NF because it has the transitive dependency of:
 - Author_ID --> Author
 - Author --> Author_Nationality
 - Therefore Author_ID --> Author_Nationality

3RD NORMAL FORM EXAMPLE

- Instead, we can remove the transitive properties by organizing the data into the tables below:

Author_ID	Author
A1	Andrew
A2	Andres
A3	Rodney

Author	Author_Nationality
Andrew	USA
Andres	USA
Rodney	Canada

ACTIVITY

- Take the following table and apply normalization so that the table is in 3rd normal form:

Student_ID	Student_Name	Address	Phone	Class_ID	Subject
1	Mary Jane	123 Main St.	1592361489, 1231231236	548	Math
2	Peter Parker	214 Bordfish Rd.	4784694528	261, 584	Science, History
3	Norman Osborn	94 Cameron St.	5686917562	885	Chemistry
4	Curtis Connors	44 Pink Ln.	1345879265	199	Herpetology